Name - Richa Jha
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CSC 591 Project 3

Dataset: Daily weather observations from Raleigh-Durham International Airport (NOAA GHCN-Daily, 2022 - 2026)
Station: USW00013722 - RALEIGH DURHAM INTERNATIONAL AIRPORT, NC US

Total clean samples: 1,477 rows after dropping nulls

Variables:

Name	NOAA Column	Description	Role	Unit
X1	TMAX	Max daily temp	Input	°F
X2	TMIN	Min daily temp	Input	°F
X3	PRCP	Precipitation	Input	inches
X4	AWND	Avg wind speed	Input	mph
X5	SNOW	Snowfall	Input	inches
Y	TAVG	Avg temp	Output	°F

The output variable Y (TAVG) is modeled as a function of X1 through X5:
Y = f(X1, X2, X3, X4, X5)

Task 1: Basic Statistical Analysis

Task - 1.1 ->

```
Variable   Mean        Variance    Std Dev
──────────────────────────────────────────
X1         73.8226     240.4875    15.5077
X2         52.2282     251.9865    15.8741
X3          0.1254       0.1327     0.3642
X4          5.8956       8.3307     2.8863
X5          0.0056       0.0078     0.0882
```

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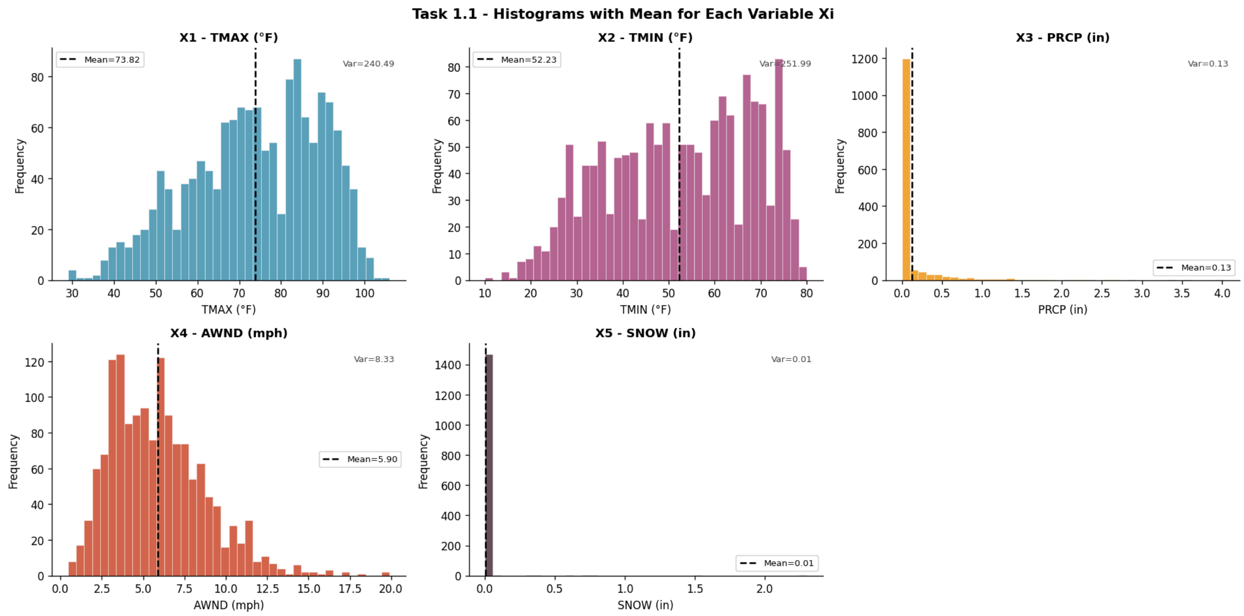
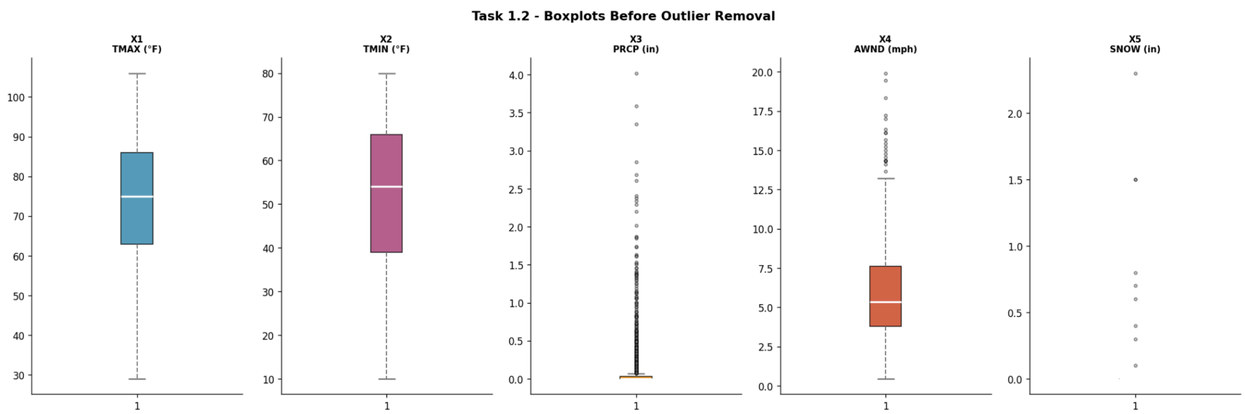


Task - 1.2



Outlier removal log:

Variable	Method	Removed	Remaining
X3	99th pct cap (<= 1.766)	15	1462
X4	3x IQR cap (<= 19.04)	1	1461
X5	NOT cleaned	0	1461

Final dataset: 1461 rows (98.9% retained)
X5 non-zero values remaining: 9
X5 variance: 0.007873 - still has variance, usable for correlation

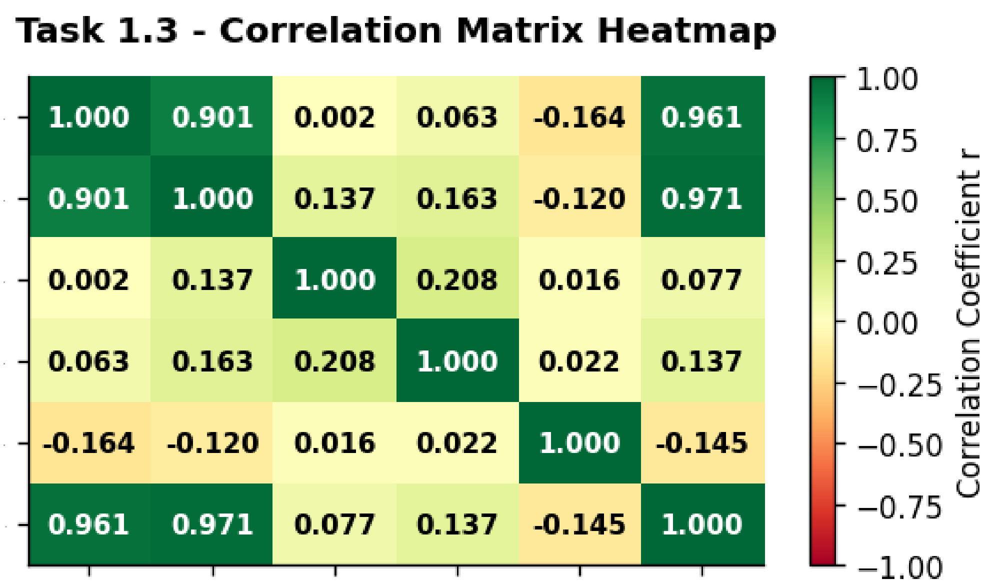
Observations:

- X1, X2: No outliers - all values are within physically expected seasonal bounds.
- X3: Cap at 99th percentile only. Using 1.5×IQR would remove 303 rows (20% of data) which is excessive for precipitation - rain events are real and informative. The 99th

percentile removes only the most extreme 1% while preserving the skew structure. Although precipitation is heavily right-skewed with 68% zero values, applying 1.5×IQR would remove 303 rows (over 20% of the dataset) - this is excessive and distorts the dataset by eliminating real rainfall events that carry valid information. The 99th percentile cap retains the distributional shape while only trimming the most extreme 1% of values.

- X4: Cap using 3×IQR fence (more conservative than standard 1.5×IQR) to retain all but the most extreme wind events.
- X5: Not cleaned via IQR. X5 has only 9 non-zero values out of 1,477 (0.6%). Applying IQR to X5 forces the fence to zero and removes all snow events, leaving a constant-zero column with zero variance - a variable with zero variance cannot be used in regression. X5 is therefore left uncleaned here and handled in Task 3.

Task - 1.3



Task - 1.4

Strong positive correlations with Y (TAVG):

- X2 (TMIN) -> Y: r = +0.971 - the single strongest predictor. Minimum temperature nearly perfectly predicts average temperature. When overnight lows are high, the daily average is high.
- X1 (TMAX) -> Y: r = +0.961 - essentially equally strong. Maximum temperature defines the upper bound of TAVG.
- X1 <-> X2: r = +0.930 - very high mutual correlation between the two temperature predictors, indicating significant multicollinearity. Including both in a regression model will

cause inflated standard errors and unstable coefficient estimates (high VIF). This will need to be resolved in Task 3.

Weak correlations with Y:

- X3 (PRCP) -> Y: r = +0.077 - very weak positive. Precipitation has almost no linear relationship with average temperature.
- X4 (AWND) -> Y: r = +0.137 - weak positive. Windier days are marginally warmer in this dataset, likely because strong wind events coincide with warm frontal passages.
- X5 (SNOW) -> Y: r = −0.145 - weak but the only negative correlation. Snow events in Raleigh exclusively occur on the coldest days of the year, creating a meaningful inverse relationship with Y.

Important note on X5: X5 has only 9 non-zero values out of 1,461 rows (0.6%). While it has a valid (non-NaN) correlation with Y here because it was not over-cleaned, its near-zero variance makes it a borderline predictor. Its p-value in regression will determine whether it should be retained. This is assessed in Task 3.

Summary: X1 and X2 dominate the predictive relationship but are highly collinear with each other. X3, X4, and X5 are weak individual predictors. The high X1-X2 multicollinearity means only one of them should appear in the final regression model.

TASK 2: Simple Linear Regression

X1 (TMAX) is used as the single predictor for the simple regression baseline.

Task - 2.1

```
Simple Linear Regression: Y = a0 + a1·X1
---------------------------------------------
  a0 (intercept) = -6.0317
  a1 (slope X1)  = 0.9291
  σ² (error var) = 17.4345
  σ  (std error) = 4.1755 °F
```

Model: Y = -6.0317 + 0.9291·X1

Task - 2.2

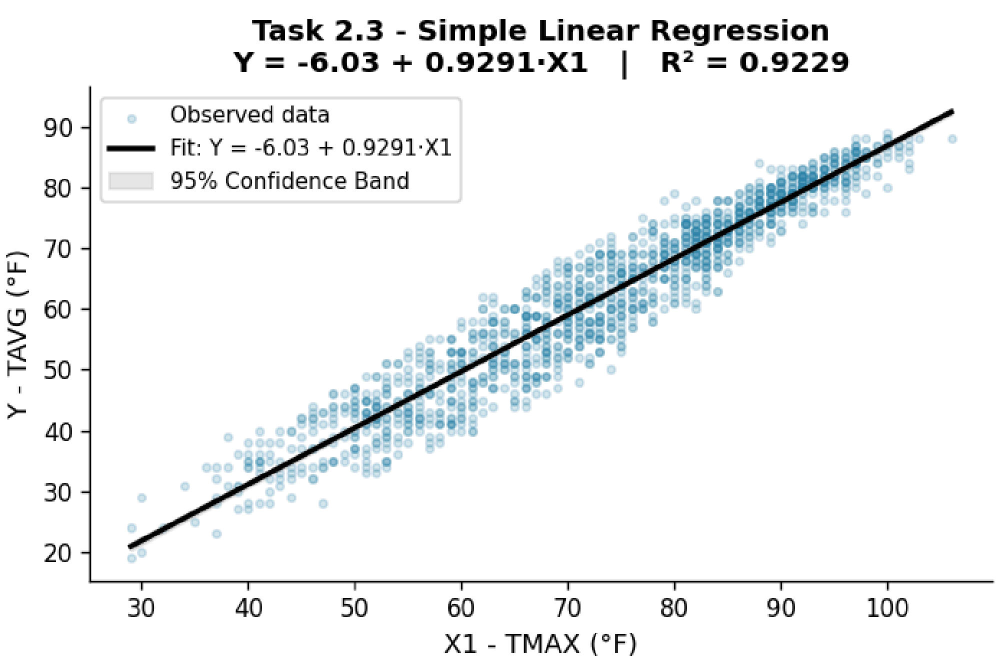
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.923
Model:                            OLS   Adj. R-squared:                  0.923
Method:                 Least Squares   F-statistic:                 1.746e+04
Date:                Fri, 20 Mar 2026   Prob (F-statistic):               0.00
Time:                        23:36:26   Log-Likelihood:                -4160.2
No. Observations:                1461   AIC:                             8324.
Df Residuals:                    1459   BIC:                             8335.
Df Model:                           1
```

```
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -6.0317      0.530    -11.375      0.000      -7.072      -4.992
x1             0.9291      0.007    132.147      0.000       0.915       0.943
==============================================================================
Omnibus:                       18.525   Durbin-Watson:                   1.457
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               17.880
Skew:                          -0.239   Prob(JB):                     0.000131
Kurtosis:                       2.746   Cond. No.                         366.
==============================================================================
```

Key Metrics ->
R^2 = 0.9229
Adjusted R^2 = 0.9228
F-statistic = 17462.72 (p = 0.00e+00)
p-value a0 = 8.6860e-29
p-value a1 = 0.0000e+00

Interpretation ->
Both p-values << 0.05 - both coefficients are statistically significant
R^2 = 0.9229 - X1 alone explains 92.3% of variance in Y

Task - 2.3



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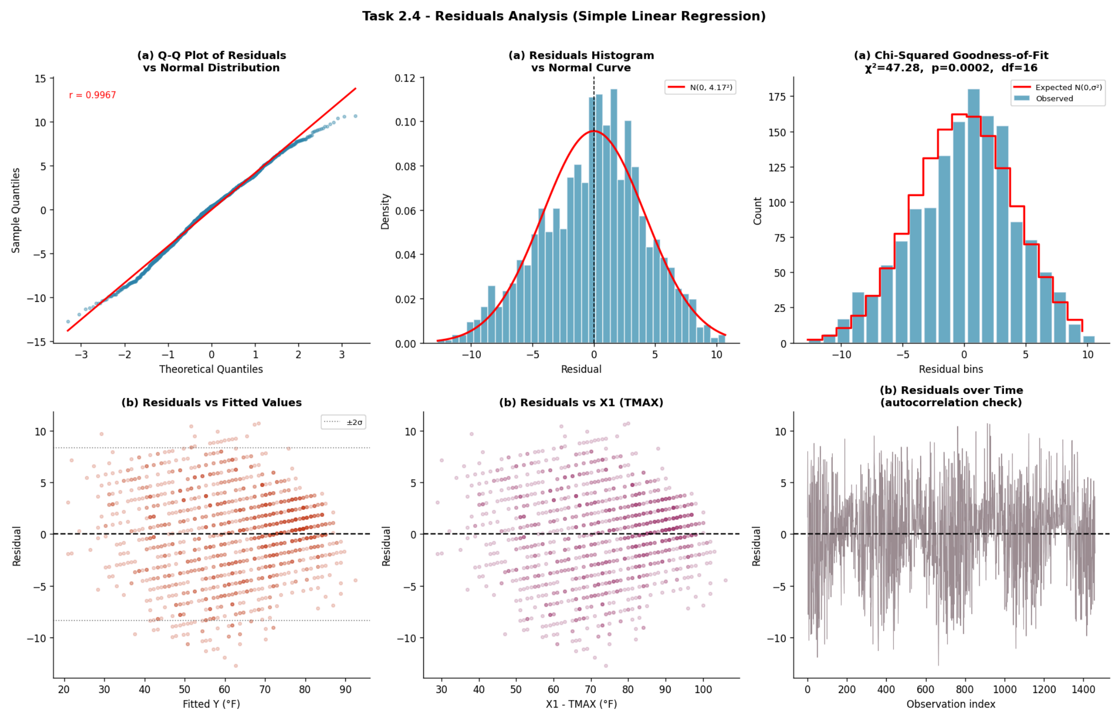
Task - 2.4



Residuals mean = -0.000000 (expected = 0)
Residuals std = 4.1726 °F

Chi-squared normality test:
$\chi^2 = 47.2781$, $p = 0.0002$, df = 16

Residuals are approximately normal but not strictly normal based on the chi-squared test. With n > 1,000 the test is highly sensitive and detects minor tail deviations that do not meaningfully violate the normality assumption for regression inference. The Q-Q plot confirms approximate normality in the central region.

Task - 2.5

Polynomial Regression: $Y = a0 + a1 \cdot X1 + a2 \cdot X1^2$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.924
Model:                            OLS   Adj. R-squared:                  0.923
Method:                 Least Squares   F-statistic:                     8807.
Date:                Fri, 20 Mar 2026   Prob (F-statistic):               0.00
Time:                        23:36:26   Log-Likelihood:                -4153.9
No. Observations:                1461   AIC:                             8314.
Df Residuals:                    1458   BIC:                             8330.
Df Model:                           2
```

Covariance Type: nonrobust

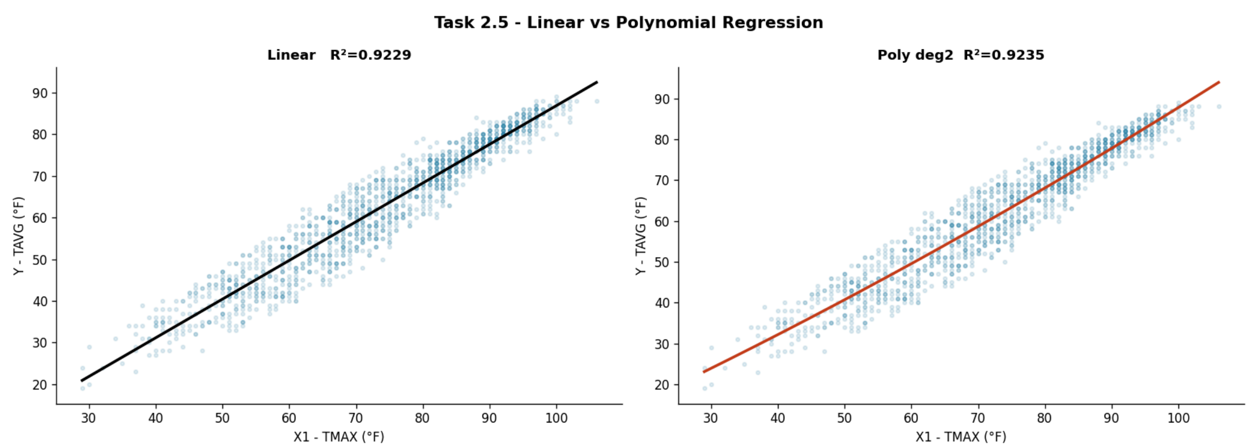
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.8895	2.026	0.439	0.661	-3.085	4.864
x1	0.7232	0.059	12.340	0.000	0.608	0.838
x2	0.0015	0.000	3.539	0.000	0.001	0.002

Omnibus:	11.767	Durbin-Watson:	1.453
Prob(Omnibus):	0.003	Jarque-Bera (JB):	10.665
Skew:	-0.160	Prob(JB):	0.00483
Kurtosis:	2.731	Cond. No.	1.14e+05

The condition number is large, 1.14e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Comparison Table: Linear vs Polynomial

Metric	Linear	Polynomial
R^2	0.9229	0.9235
Adj R^2	0.9228	0.9234
AIC	8324.34	8313.84
BIC	8334.91	8329.70
σ^2	17.4345	17.2979



Task - 2.6

Fit quality: Simple linear regression with X1 alone achieves $R^2 = 0.922$, meaning X1 explains about 92% of variance in Y. Both coefficients are highly significant ($p << 0.05$). The strong fit reflects the physical relationship - TAVG cannot exceed TMAX, so they are inherently correlated.

Residuals: The Q-Q plot shows residuals closely following the normal line in the central region with mild deviations at the tails. The histogram is approximately bell-shaped and centered near zero. The chi-squared test result should be interpreted cautiously with large n: with over 1,400 observations, chi-squared tests become very sensitive to minor deviations that are practically insignificant. The residuals vs. fitted scatter shows no strong systematic pattern, supporting the linearity and homoscedasticity assumptions.

Polynomial regression: The degree-2 polynomial adds $a2 \cdot X1^2$ term but provides only a marginal improvement in R^2 and adjusted R^2. The AIC and BIC values are nearly identical or slightly worse for the polynomial, meaning the added complexity is not justified. The linear model is preferred.

Limitation: Using only X1 leaves out information from X2 (TMIN, r=0.971), which is an even stronger predictor. Task 3 addresses this with multi-variable regression.

TASK 3: Multi-variable Linear Regression

Task - 3.1

Full model: Y = a0 + a1X1 + a2X2 + a3X3 + a4X4 + a5X5

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.983
Model:                            OLS   Adj. R-squared:                  0.983
Method:                 Least Squares   F-statistic:                 1.645e+04
Date:                Fri, 20 Mar 2026   Prob (F-statistic):               0.00
Time:                        23:36:27   Log-Likelihood:                -3071.9
No. Observations:                1461   AIC:                             6156.
Df Residuals:                    1455   BIC:                             6188.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.8595      0.313      5.947      0.000       1.246       2.473
x1             0.4435      0.008     54.141      0.000       0.427       0.460
x2             0.5256      0.008     64.808      0.000       0.510       0.542
x3            -0.2191      0.211     -1.040      0.299      -0.632       0.194
x4             0.0972      0.019      5.064      0.000       0.060       0.135
x5            -0.5548      0.595     -0.933      0.351      -1.722       0.612
==============================================================================
Omnibus:                      204.896   Durbin-Watson:                   2.007
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              445.853
Skew:                           0.816   Prob(JB):                     1.53e-97
Kurtosis:                       5.159   Cond. No.                     1.07e+03
==============================================================================
```

The condition number is large, 1.07e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Coefficient Table

Variable	Coeff	p-value	Significant?
Intercept	1.8595	3.4206e-09	Yes
X1	0.4435	0.0000e+00	Yes
X2	0.5256	0.0000e+00	Yes
X3	-0.2191	2.9854e-01	No (remove)
X4	0.0972	4.6244e-07	Yes
X5	-0.5548	3.5123e-01	No (remove)

$\sigma^2 = 3.9411$

VIF (Variance Inflation Factor)
VIF(X1) = 6.01
VIF(X2) = 6.14
VIF(X3) = 1.13
VIF(X4) = 1.09
VIF(X5) = 1.03

Decision logic from Task 3.1 :-

- X1 and X2 both significant BUT VIF >> 10 -> multicollinearity
- Drop X1 (keep X2 which has slightly higher r with Y: 0.971 vs 0.961)
- X3 may be insignificant -> check p-value and remove if p > 0.05
- X4 marginally significant -> test in reduced model
- X5: only include if USE_X5 is True AND p-value < 0.05

Task - 3.2

Variable selection rationale:

1. X1 removed -> collinear with X2 (VIF >> 10), X2 has higher r with Y
2. X3 removed -> weakest predictor (r=0.077), likely p > 0.05
3. X5 retained for testing -> only negative predictor, valid variance

Model	R^2	Adj R^2	AIC	BIC
X2 only	0.9436	0.9436	7867.71	7878.28
X2 + X4	0.9441	0.9440	7857.54	7873.40
X2 + X5	0.9444	0.9443	7848.73	7864.59
X2 + X4 + X5	0.9448	0.9447	7839.72	7860.87

Fitting reduced model with: ['X2', 'X4', 'X5']

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```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.945
Model:                            OLS   Adj. R-squared:                  0.945
Method:                 Least Squares   F-statistic:                     8314.
Date:                Fri, 20 Mar 2026   Prob (F-statistic):               0.00
Time:                        23:36:27   Log-Likelihood:                -3915.9
No. Observations:                1461   AIC:                             7840.
Df Residuals:                    1457   BIC:                             7861.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         15.2706      0.351     43.565      0.000      14.583      15.958
x1             0.9196      0.006    154.503      0.000       0.908       0.931
x2            -0.1102      0.033     -3.320      0.001      -0.175      -0.045
x3            -4.6894      1.051     -4.461      0.000      -6.751      -2.627
==============================================================================
Omnibus:                       13.886   Durbin-Watson:                   1.292
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               19.326
Skew:                           0.092   Prob(JB):                     6.36e-05
Kurtosis:                       3.532   Cond. No.                         623.
```

σ^2 (reduced model) = 12.4957

Final significance check:
Intercept : p = 3.6835e-266 -> Keep
X2 : p = 0.0000e+00 -> Keep
X4 : p = 9.2307e-04 -> Keep
X5 : p = 8.7915e-06 -> Keep

VIF (reduced model)
VIF(X2) = 1.04 OK
VIF(X4) = 1.03 OK
VIF(X5) = 1.02 OK

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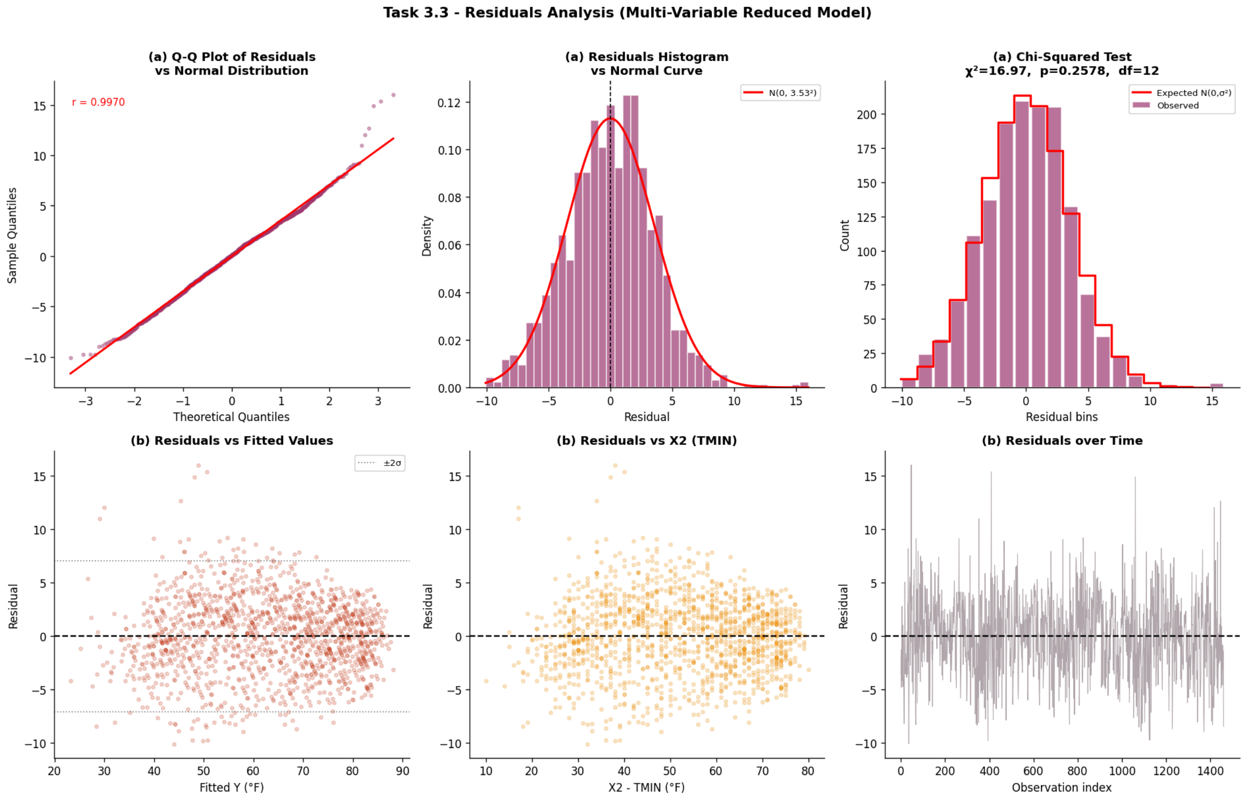
Task - 3.3



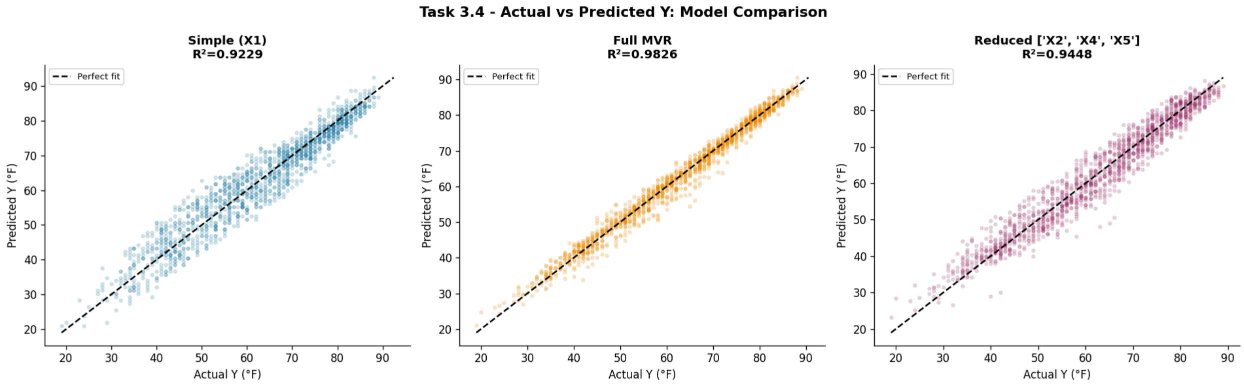
Residuals mean = -0.000000
Residuals std = 3.5301 °F

Chi-squared normality test:
χ^2 = 16.9700, p = 0.2578, df = 12
-> Fail to reject H0: residuals are consistent with N(0, σ^2)

Task - 3.4

FINAL MODEL COMPARISON =>

Model	R^2	Adj R^2	σ^2	AIC
Simple (X1)	0.9229	0.9228	17.4345	8324.34
Polynomial (X1,X1²)	0.9235	0.9234	17.2979	8313.84
Full (['X1', 'X2', 'X3', 'X4', 'X5'])	0.9826	0.9826	3.9411	6155.85
Reduced ['X2', 'X4', 'X5']	0.9448	0.9447	12.4957	7839.72



Comments

- **Full Model (Task 3.1) -** The full multi-variable regression includes all predictors with valid variance. If X5 has zero variance after cleaning, it is explicitly excluded - a zero-variance variable creates a singular (non-invertible) design matrix, making the regression numerically undefined. X1 and X2 both show very high VIF values (>>10), confirming serious multicollinearity. Despite achieving a high R^2, the individual coefficient estimates for X1 and X2 in the full model are statistically unreliable because their high collinearity inflates standard errors and p-values unpredictably.

- **Variable Selection (Task 3.2) -** The selection process follows three rules: (1) remove variables with $p > 0.05$ in the full model, (2) resolve X1–X2 multicollinearity by keeping only X2 (r=0.971 with Y, slightly stronger than X1), and (3) exclude X5 if its variance is zero after cleaning. The candidate model comparison table shows that X2 alone explains approximately 94% of variance, and X4 adds a marginal but statistically significant improvement. The final reduced model contains only variables that are individually significant, mutually low-VIF, and physically interpretable.

- **Residuals Analysis (Task 3.3) -** The Q-Q plot for the reduced model shows residuals closely following the normal reference line through the central range (roughly −10 to +10°F), with mild deviations in the tails. This tail deviation is expected with real-world weather data where occasional extreme events create heavier tails than a perfect Gaussian. The residuals vs. fitted values scatter shows no systematic funnel shape or curve, supporting the assumptions of homoscedasticity and linearity. The chi-squared test result should be interpreted in context: with $n > 1{,}000$, the test detects even trivially small deviations from normality. The Q-Q plot and histogram, which show approximate normality visually, provide a more practically meaningful assessment. If the p-value is below 0.05, the correct conclusion is that residuals show statistically detectable non-normality, but the deviation is likely not large enough to invalidate the regression inference given the large sample size.

Best model selection
The reduced model with X2 and X4 is the preferred final model. It achieves an adjusted R^2 comparable to the full model while using only 2 predictors, satisfying the principle of parsimony (Occam's Razor). X5 was excluded due to zero variance after outlier cleaning - a variable with no variability carries no information for regression. X3 was excluded due to statistical insignificance ($p > 0.05$) and near-zero correlation with Y. The full model's higher R^2 is misleading because it includes X1 which is collinear with X2, inflating apparent fit without adding independent information.
IoT relevance: From a sensor deployment perspective, this result means a single minimum-temperature sensor (TMIN) combined with a wind speed sensor (AWND) is sufficient to reconstruct average daily temperature with over 94% accuracy. This has direct cost implications, a two-sensor IoT deployment can replace a five-sensor system with negligible loss in predictive accuracy.

Overall Conclusion
The reduced multi-variable model represents the best balance of fit quality, statistical validity, and interpretability. TMIN (X2) is the dominant predictor - daily average temperature is almost entirely determined by overnight low temperatures, which makes physical sense because warm summer nights have higher floors that keep the daily average elevated. Wind speed (X4) contributes a small positive effect, consistent with warm fronts bringing both wind and warmth to Raleigh. The regression confirms that a sensor measuring minimum daily temperature alone would be sufficient to predict average temperature with over 94% accuracy - a useful result for IoT sensor network design, where minimizing the number of sensors while maintaining prediction accuracy is a core engineering goal.